

PROJEKT NA ZALICZENIE LABORATORIÓW Z LINUXA

Instalacja i Konfiguracja Kubernetesa

PLAN

- Faza 1: Przygotowanie warstwy fizycznej i sieciowej
- Faza 2: Konfiguracja systemów i instalacja usług
- Faza 3: Budowa klastra K3s
- Faza 4: Wdrożenie aplikacji na K3s
- Faza 5: Wdrożenie ArgoCD

KUBERNETES (K8S) VS K3S – ORKIESTRACJA W SKALI MIKRO

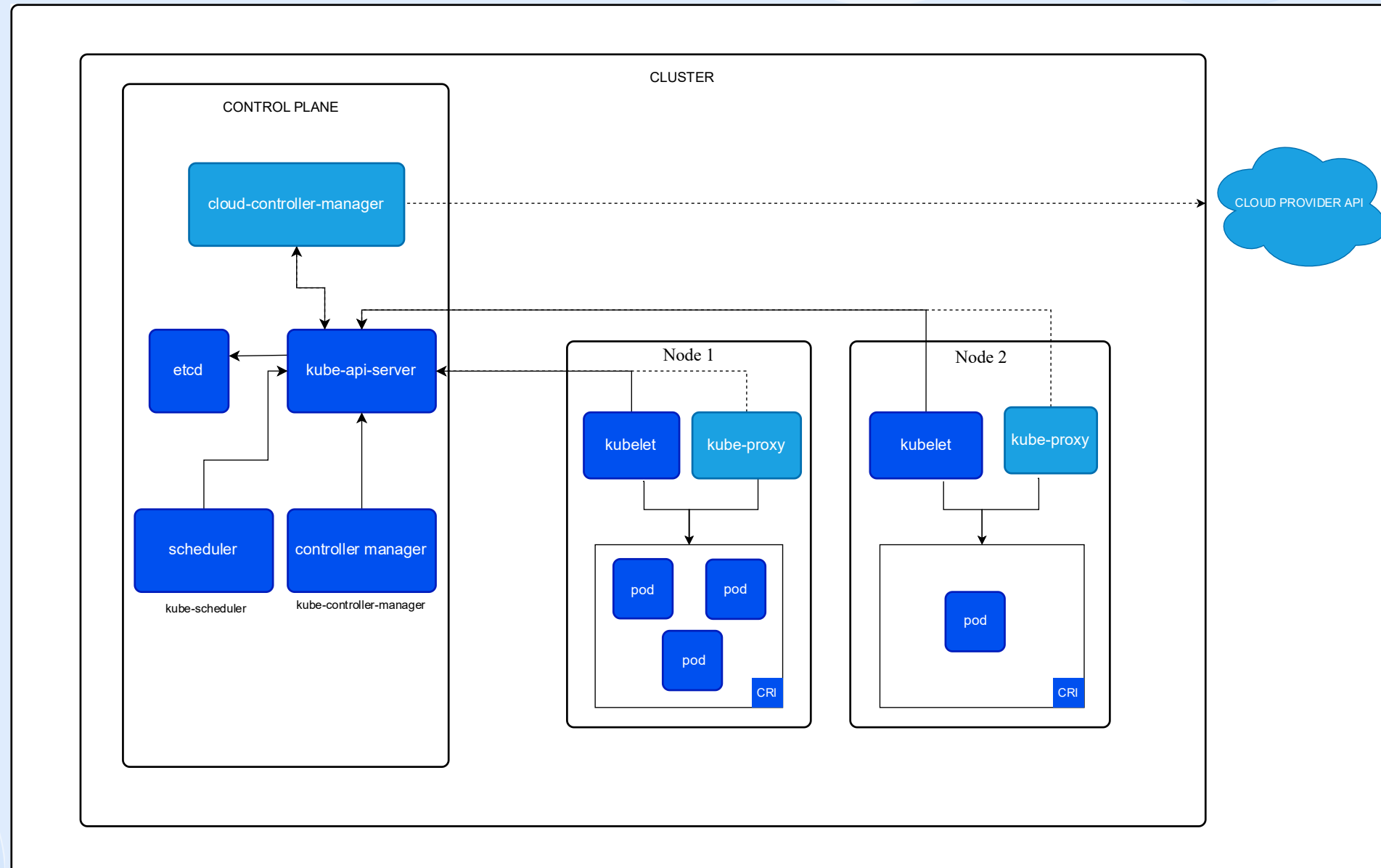
- **Kubernetes (K8s):** System do automatyzacji wdrażania, skalowania i zarządzania skonteneryzowanymi aplikacjami w klastrach.
- **K3s:** Lekka, w pełni certyfikowana dystrybucja Kubernetes od firmy Rancher.

BUDOWA KLASTRA K3S

- **Node 1 (EndeavourOS):** Control Plane zarządzający klastrem.
- **Node 2 (Debian 12):** Klasyczny worker z natywną bazą MariaDB na hoście sterowaną przez **systemd**.
- **Node 3 (Flatcar):** Niezmienny (Immutable OS) system uruchomieniowy dla aplikacji produkcyjnej.



ARCHITEKTURA K8S



ANATOMIA WDROŻENIA APLIKACJI (OBIEKTY K3S)

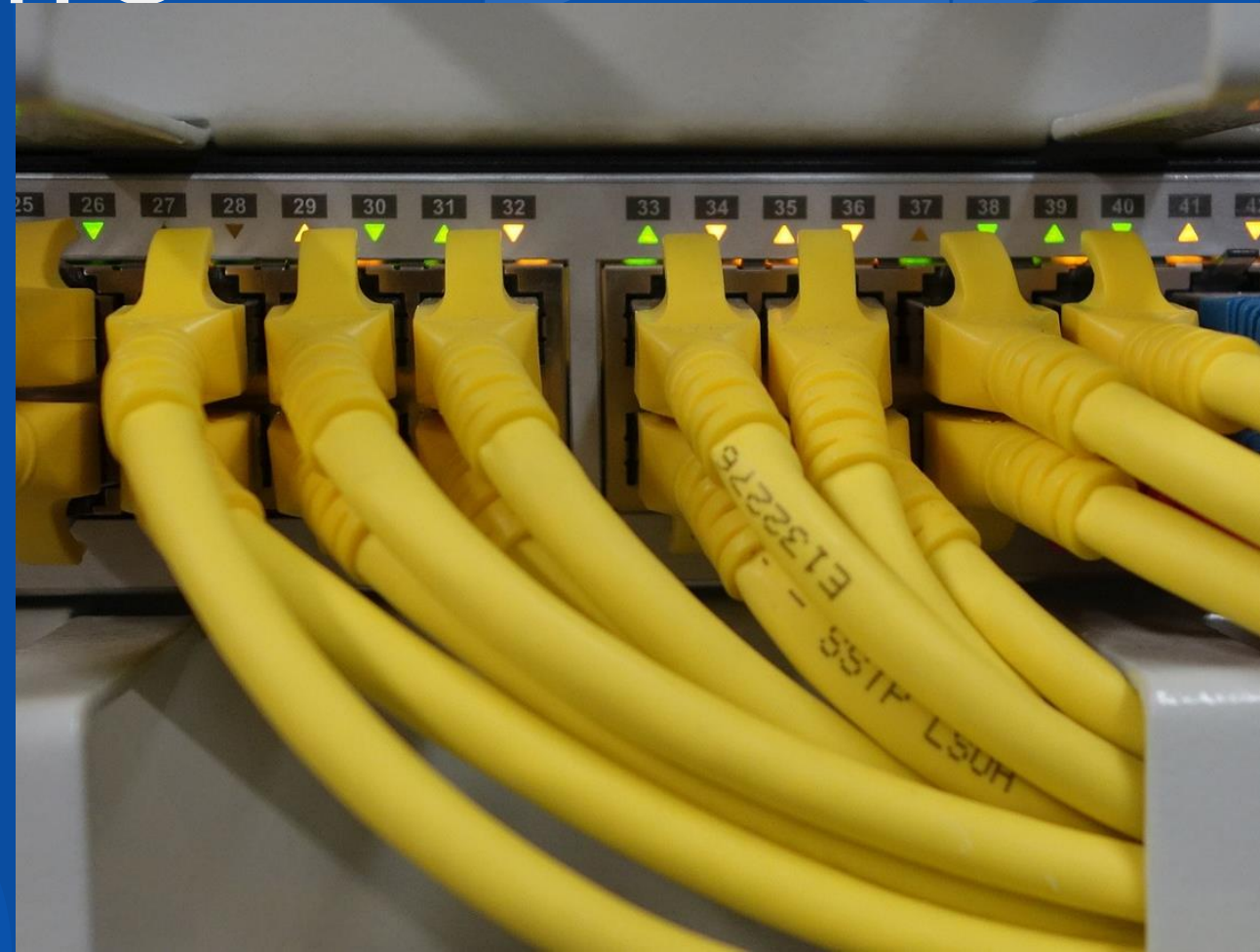
- **Namespace (**prod**)**: Logiczna izolacja zasobów produkcyjnych od deweloperskich w jednym klastrze.
- **ConfigMap & Secret**: Separacja konfiguracji (np. adres IP bazy) i danych poufnych (hasła) od samego kodu aplikacji.
- **PersistentVolumeClaim (PVC)**: Żądanie przydzielenia trwałej pamięci dyskowej na pliki wgrywane przez użytkowników (używamy Local Path Provisionera).
- **Deployment**: Deklaratywny opis stanu pożądanego aplikacji (liczba replik, limity zasobów CPU/RAM, sondy żywotności **livenessProbe** i gotowości **readinessProbe**).
- **Service (ClusterIP)**: Wewnętrzny load balancer dający stały adres IP i nazwę DNS dla podów aplikacji.
- **Ingress**: Reguła routingu wystawiająca aplikację na świat zewnętrzny przez kontroler Ingress (Traefik).
- **NetworkPolicy**: Firewall wewnątrzklastrowy, który blokuje cały ruch do aplikacji, zezwalając na dostęp wyłącznie z kontrolera Ingress (przestrzeń **kube-system**).
- **HorizontalPodAutoscaler (HPA)**: Automatyczne skalowanie liczby podów od 2 do 5 w zależności od obciążenia procesora (próg: 80% CPU).

GITOPS ORAZ ARGOCD – INFRASTRUKTURA JAKO KOD W PRAKTYCE

- **Definicja GitOps:** Praktyka operacyjna, w której repozytorium Git stanowi **jedyne źródło prawdy** (Single Source of Truth) o stanie całej infrastruktury.
- **Rola ArgoCD:** Kontroler działający wewnątrz klastra K3s, który w pętli (Pull Model) porównuje pliki YAML z Git ze stanem rzeczywistym w klastrze.
- **Zalety rozwiązania:**
 - Brak konieczności ręcznego używania **kubectl apply**.
 - W przypadku awarii klastra, odtworzenie całego środowiska produkcyjnego to kwestia podpięcia ArgoCD pod repozytorium Git.
 - Automatyczne cofanie ręcznych (nieautoryzowanych) zmian w klastrze (Self-healing / Drift Detection).

FAZA 1

Przygotowanie warstwy fizycznej i sieciowej



CELE

- Instalacja systemów operacyjnych
- Ustawienie stałych adresów IP na systemach
- Wdrożenie logowania tylko przez klucz SSH
- Skonfigurowanie Firewalli
- Stworzenie pliku ignition.json potrzebnego do instalacji Flatcara

Konfiguracja sieciowa na EndeavourOS

```
~ » sudo vim /etc/NetworkManager/system-connections/Orange_Swiatlowod_D760.nmconnection
```

```
~ took 1m9s
```

```
~ » sudo nmcli connection reload
```

```
~
```

```
~ » sudo nmcli connection up "Orange_Swiatlowod_D760"
```

```
Pomyślnie aktywowano połączenie (ścieżka aktywacji D-Bus: /org/freedesktop/NetworkManager/ActiveConnection/18)
```

```
[connection]
```

```
id=Orange_Swiatlowod_D760
```

```
uuid=c1425e0c-d0fa-4e20-bddf-c475074b6c81
```

```
type=wifi
```

```
[wifi]
```

```
mode=infrastructure
```

```
ssid=Orange_Swiatlowod_D760
```

```
[wifi-security]
```

```
key-mgmt=wpa-psk
```

```
psk=Ag9S5AnLtpWGMf4dQX
```

```
[ipv4]
```

```
address1=192.168.8.150/24,192.168.8.1
```

```
dns=1.1.1.1;8.8.8.8;
```

```
method=manual
```

```
[ipv6]
```

```
addr-gen-mode=stable-privacy
```

```
method=ignore
```

```
[proxy]
```

```
~ » sudo firewall-cmd --list-ports
```

```
6443/tcp 10250/tcp 8472/udp
```

```
~/Pulpit/Projekt Kubernetes
```

```
~ » sudo firewall-cmd --permanent --add-port=80/tcp
```

```
[sudo] hasło użytkownika mpolak:
```

```
success
```

```
~/Pulpit/Projekt Kubernetes took 6s
```

```
~ » sudo firewall-cmd --permanent --add-port=443/tcp
```

```
success
```

```
~/Pulpit/Projekt Kubernetes
```

```
~ » sudo firewall-cmd --reload
```

```
success
```

```
~/Pulpit/Projekt Kubernetes took 2s
```

```
~ » sudo firewall-cmd --permanent --zone=trusted --add-source=10.42.0.0/16
```

```
[sudo] hasło użytkownika mpolak:
```

```
success
```

```
~/Pulpit/Projekt Kubernetes took 6s
```

```
~ » sudo firewall-cmd --permanent --zone=trusted --add-source=10.43.0.0/16
```

```
success
```

```
~/Pulpit/Projekt Kubernetes
```

```
~ » sudo firewall-cmd --reload
```

```
success
```

Konfiguracja sieciowa na Debianie

```
mpolak@old-dell:~$ sudo systemctl enable --now systemd-resolved
mpolak@old-dell:~$ sudo ln -sf /run/systemd/resolve/stub-resolv.conf /etc/resolv.conf
mpolak@old-dell:~$ sudo vim /etc/network/interfaces
mpolak@old-dell:~$ sudo systemctl enable --now systemd-networkd
Created symlink /etc/systemd/system/dbus-org.freedesktop.network1.service → /lib/systemd/system/systemd-networkd.service.
Created symlink /etc/systemd/system/multi-user.target.wants/systemd-networkd.service → /lib/systemd/system/systemd-networkd.service.
Created symlink /etc/systemd/system/sockets.target.wants/systemd-networkd.socket → /lib/systemd/system/systemd-networkd.socket.
Created symlink /etc/systemd/system/sysinit.target.wants/systemd-network-generator.service → /lib/systemd/system/systemd-network-generator.service.
Created symlink /etc/systemd/system/network-online.target.wants/systemd-networkd-wait-online.service → /lib/systemd/system/systemd-networkd-wait-online.service.
Connection to 192.168.8.45 closed.
debug1: channel 0: free: client-session, nchannels 1
Transferred: sent 9804, received 51504 bytes, in 4309.9 seconds
Bytes per second: sent 2.3, received 12.0
debug1: Exit status -1
mpolak@old-dell:~$ networkctl status
●      State: routable
  Online state: online
    Address: 192.168.8.160 on wlp2s0
              fe80::a11:96ff:fe0:73bc on wlp2s0
    Gateway: 192.168.8.1 on wlp2s0
      DNS: 1.1.1.1
           8.8.8.8
mpolak@old-dell:~$ ip -br -4 a
lo                UNKNOWN        127.0.0.1/8
wlp2s0            UP                192.168.8.160/24
mpolak@old-dell:~$ ping -c 4 google.com
PING google.com (142.250.120.101) 56(84) bytes of data:
64 bytes from zo-in-f101.1e100.net (142.250.120.101): icmp_seq=1 ttl=115 time=10.1 ms
64 bytes from zo-in-f101.1e100.net (142.250.120.101): icmp_seq=2 ttl=115 time=9.04 ms
64 bytes from zo-in-f101.1e100.net (142.250.120.101): icmp_seq=3 ttl=115 time=8.24 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 8.236/9.133/10.125/0.774 ms
```

```
[Match]
Name=wlp2s0

[Network]
Address=192.168.8.160/24
Gateway=192.168.8.1
DNS=1.1.1.1
DNS=8.8.8.8
```

Konfiguracja firewalla (UFW) na Debianie

```
mpolak@old-dell:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
mpolak@old-dell:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
mpolak@old-dell:~$ ^C
mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 22 proto tcp
Rules updated
mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 3306 proto tcp
Rules updated
mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 10250 proto tcp
Rules updated
mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 30000:32767 proto tcp
Rules updated
mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 30000:32767 proto udp
Rules updated
mpolak@old-dell:~$ sudo vim /etc/default/ufw
mpolak@old-dell:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
mpolak@old-dell:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN 192.168.8.0/24
3306/tcp ALLOW IN 192.168.8.0/24
10250/tcp ALLOW IN 192.168.8.0/24
30000:32767/tcp ALLOW IN 192.168.8.0/24
30000:32767/udp ALLOW IN 192.168.8.0/24

mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 8472 proto udp
Rule added
mpolak@old-dell:~$ sudo ufw allow from 192.168.8.0/24 to any port 80,443 proto tcp
Rule added
mpolak@old-dell:~$ sudo ufw allow from 10.42.0.0/16 to any port 3306 proto tcp
Rule added
mpolak@old-dell:~$ sudo ufw allow in on cni0
Rule added
Rule added (v6)
mpolak@old-dell:~$ sudo ufw allow in on flannel.1
Rule added
Rule added (v6)
mpolak@old-dell:~$ sudo ufw reload
Firewall reloaded
```

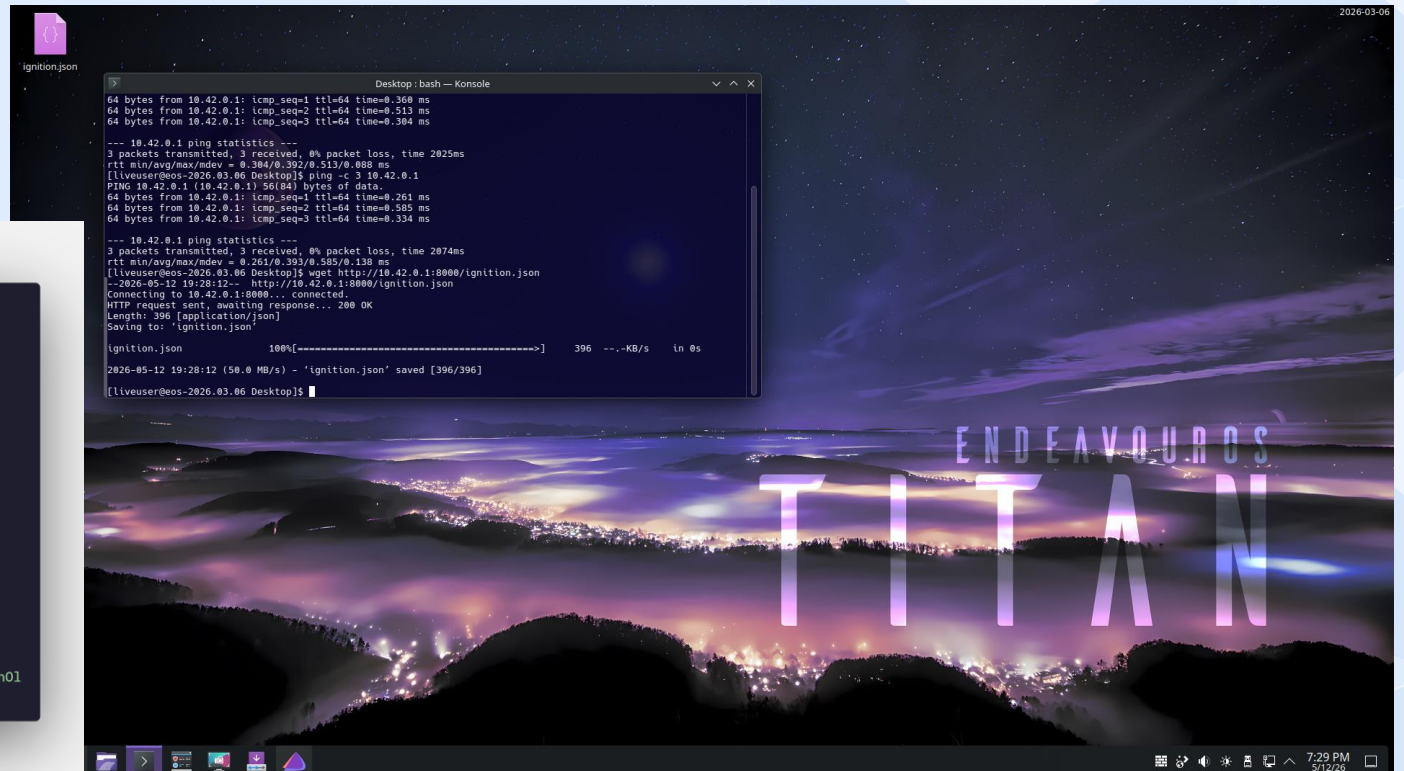
Instalacja Flatcar

```
variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/systemd/network/10-static.network
      contents:
        inline: |
          [Match]
          Name=e*

          [Network]
          Address=192.168.8.170/24
          Gateway=192.168.8.1
          DNS=1.1.1.1
          DNS=8.8.8.8
passwd:
  users:
    - name: core
      ssh_authorized_keys:
        - "ssh-ed25519 AAAAC3NzaC11ZD11NTE5AAAAID5NRbpq4HgJ1BXoJ16EZReoh01
        UMBUMxb2z+ECRt1m Endeavour-to-Flatcar"
```

```
{
  "ignition": {
    "version": "3.3.0"
  },
  "passwd": {
    "users": [
      {
        "name": "core",
        "sshAuthorizedKeys": [
          "ssh-ed25519 AAAAC3NzaC11ZD11NTE5AAAAID5NRbpq4HgJ1BXoJ16EZReoh01
          UMBUMxb2z+ECRt1m Endeavour-to-Flatcar"
        ]
      }
    ]
  },
  "storage": {
    "files": [
      {
        "path": "/etc/systemd/network/10-static.network",
        "contents": {
          "compression": "",
          "source": ""
        }
      }
    ]
  },
  "data": {
    [Match]
    Name=e*

    [Network]
    Address=192.168.8.170/24
    Gateway=192.168.8.1
    DNS=1.1.1.1
    DNS=8.8.8.8
  }
}
```



```
[liveuser@eos-2026.03.06 Desktop]$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 3G 1 loop /run/archiso/airootfs
sda 8:0 0 119.2G 0 disk
├─sda1 8:1 0 50M 0 part
├─sda2 8:2 0 118.7G 0 part
├─sda3 8:3 0 537M 0 part
sdb 8:16 1 7.8G 0 disk
├─sdb1 8:17 1 3.7G 0 part /run/archiso/bootmnt
└─sdb2 8:18 1 251M 0 part

[liveuser@eos-2026.03.06 Desktop]$ wget https://raw.githubusercontent.com/flatcar/init/flatcar-master/bin/flatcar-install
--2026-05-12 19:37:54-- https://raw.githubusercontent.com/flatcar/init/flatcar-master/bin/flatcar-install
Loaded CA certificate '/etc/ssl/certs/ca-certificates.crt'
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 40257 (39K) [text/plain]
Saving to: 'flatcar-install'

flatcar-install 100%[=====] 39.31K ---KB/s in 0.03s

2026-05-12 19:37:55 (1.36 MB/s) - 'flatcar-install' saved [40257/40257]

[liveuser@eos-2026.03.06 Desktop]$ chmod +x flatcar-install
[liveuser@eos-2026.03.06 Desktop]$ sudo ./flatcar-install -d /dev/sda -i ignition.json
Current version of Flatcar Container Linux stable is 4593.2.1
Downloading the signature for https://stable.release.flatcar-linux.net/amd64-usr/4593.2.1/flatcar_production_image.bin.bz2...
2026-05-12 19:39:04 URL:https://flatcar.cdn.cncf.io/stable/amd64-usr/4593.2.1/flatcar_production_image.bin.bz2.sig [622/622] -> "/tmp/flatcar-install.xSgfFahdg5/flatcar_production_image.bin.bz2.sig" [1]
Downloading, writing and verifying flatcar_production_image.bin.bz2...
2026-05-12 19:39:47 URL:https://flatcar.cdn.cncf.io/stable/amd64-usr/4593.2.1/flatcar_production_image.bin.bz2 [495549998/495549998] -> "-" [1]
gpg: Signature made Sat 09 May 2026 02:48:05 AM CEST
gpg: using RSA key 52F145DFD0B8BD0CD928CB85A32DA80F91EF52974
gpg: issuer "buildbot@flatcar-linux.org"
gpg: Good signature from "Flatcar Buildbot (Official Builds) <buildbot@flatcar-linux.org>" [unknown]
gpg: Signature notation: manu-2.541.11.2.2
gpg: WARNING: The key's User ID is not certified with a trusted signature!
gpg: There is no indication that the signature belongs to the owner.
Primary key fingerprint: F8BC FEDE FF29 A5B4 D952 3864 E250 9AED 0593 B34A
Subkey fingerprint: 52F1 45DF D0B8 BD0C 928C BB5A 32DA 80F9 1EF5 2974
Installing Ignition config ignition.json
Success! Flatcar Container Linux stable 4593.2.1 is installed on /dev/sda
```


FAZA 2

Konfiguracja systemów instalacja usług



CELE

- Instalacja i konfiguracja bazy danych MariaDB (zamiennik MySQL) na Debianie
- Próba połączenia GPU z K3s (nieudana -> problemem jest hybrydowa architektura GPU)

Instalacja i Konfiguracja MariaDB na Debianie #1

```
mpolak@old-dell:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-12 22:34:22 CEST; 1min 29s ago
     Docs: man:mariadbd(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 2259 (mariadbd)
   Status: "Taking your SQL requests now..."
     Tasks: 9 (limit: 29468)
    Memory: 110.9M
       CPU: 708ms
    CGroup: /system.slice/mariadb.service
            └─2259 /usr/sbin/mariadbd

May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Note] InnoDB: log sequence number 46846; transaction log
May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/
May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Note] Plugin 'FEEDBACK' is disabled.
May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Warning] You need to use --log-bin to make --expire_logs_days
May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Note] Server socket created on IP: '127.0.0.1', port: 3306
May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Note] InnoDB: Buffer pool(s) load completed at 260512223422
May 12 22:34:22 old-dell mariadbd[2259]: 2026-05-12 22:34:22 0 [Note] /usr/sbin/mariadbd: ready for connections.
May 12 22:34:22 old-dell mariadbd[2259]: Version: '10.11.14-MariaDB-0+deb12u2' socket: '/run/mysqld/mysqld.sock' port: 3306
May 12 22:34:22 old-dell systemd[1]: Started mariadb.service - MariaDB 10.11.14 database server.
May 12 22:34:22 old-dell /etc/mysql/debian-start[2275]: Upgrading MySQL tables if necessary.
mpolak@old-dell:~$ sudo mariadb-secure-installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] y
Enabled successfully!
Reloading privilege tables..
... Success!

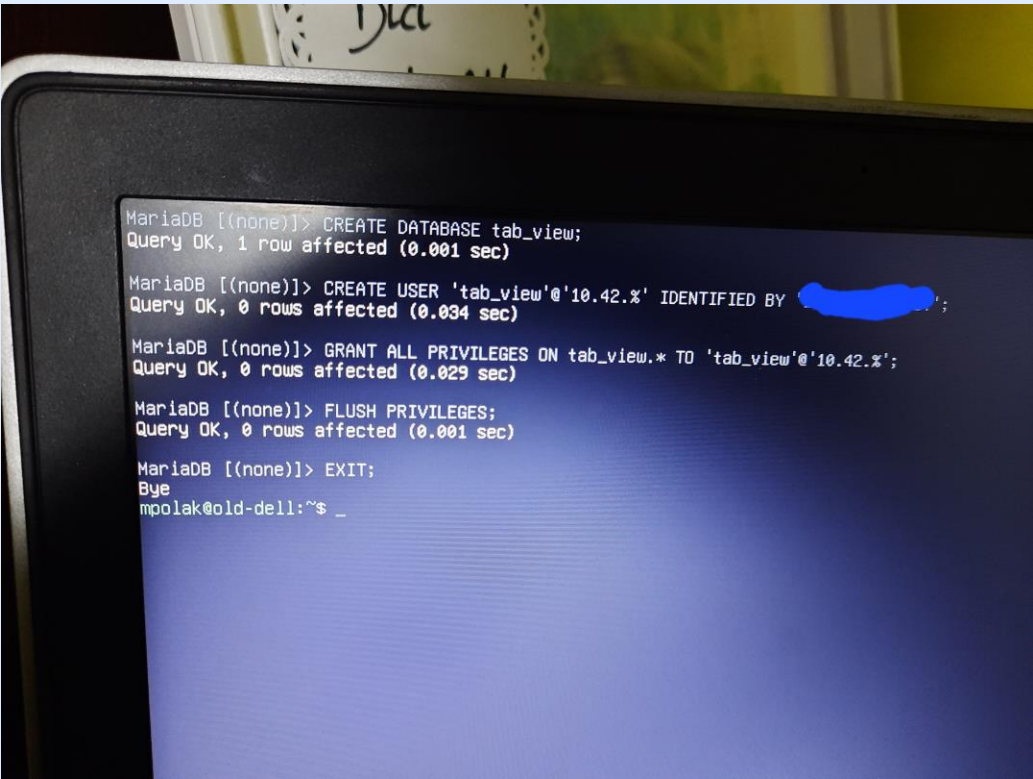
You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!
```


Instalacja i Konfiguracja MariaDB na Debianie #2

```
# Instead of skip-networking the default is now to listen only on  
# localhost which is more compatible and is not less secure.  
bind-address            = 0.0.0.0
```

```
mpolak@old-dell:~$ sudo vim /etc/mysql/mariadb.conf.d/50-server.cnf  
mpolak@old-dell:~$ sudo systemctl restart mariadb
```



```
MariaDB [(none)]> CREATE DATABASE tab_view;  
Query OK, 1 row affected (0.001 sec)  
  
MariaDB [(none)]> CREATE USER 'tab_view'@'10.42.%' IDENTIFIED BY '123456';  
Query OK, 0 rows affected (0.034 sec)  
  
MariaDB [(none)]> GRANT ALL PRIVILEGES ON tab_view.* TO 'tab_view'@'10.42.%';  
Query OK, 0 rows affected (0.029 sec)  
  
MariaDB [(none)]> FLUSH PRIVILEGES;  
Query OK, 0 rows affected (0.001 sec)  
  
MariaDB [(none)]> EXIT;  
Bye  
mpolak@old-dell:~$
```

You already have your root account protected, so you can safely answer 'n'.

```
Switch to unix_socket authentication [Y/n] y  
Enabled successfully!  
Reloading privilege tables..  
... Success!
```

You already have your root account protected, so you can safely answer 'n'.

```
Change the root password? [Y/n] y  
New password:  
Re-enter new password:  
Password updated successfully!  
Reloading privilege tables..  
... Success!
```

By default, a MariaDB installation has an anonymous user, allowing anyone to log into MariaDB without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

```
Remove anonymous users? [Y/n] y  
... Success!
```

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

```
Disallow root login remotely? [Y/n] y  
... Success!
```

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

```
Remove test database and access to it? [Y/n] y  
- Dropping test database...  
... Success!  
- Removing privileges on test database...  
... Success!
```

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

```
Reload privilege tables now? [Y/n] y  
... Success!
```

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB installation should now be secure.

Thanks for using MariaDB!

Instalacja i Konfiguracja MariaDB na Debianie #3

```
mpolak@old-dell:~$ sudo ufw allow from 10.42.0.0/16 to any port 3306 comment 'K3s Pod Network'
[sudo] password for mpolak:
Rule added
mpolak@old-dell:~$ sudo ufw allow from 192.168.1.0/24 to any port 3306 comment 'Home LAN'
Rule added
mpolak@old-dell:~$ sudo ufw status
Status: active

To                                Action      From
--                                -
22/tcp                            ALLOW       192.168.8.0/24
3306/tcp                          ALLOW       192.168.8.0/24
10250/tcp                         ALLOW       192.168.8.0/24
30000:32767/tcp                   ALLOW       192.168.8.0/24
30000:32767/udp                   ALLOW       192.168.8.0/24
8472/udp                          ALLOW       192.168.8.0/24
80,443/tcp                       ALLOW       192.168.8.0/24
3306/tcp                          ALLOW       10.42.0.0/16
Anywhere on cni0                  ALLOW       Anywhere
Anywhere on flannel.1             ALLOW       Anywhere
3306                              ALLOW       10.42.0.0/16          # K3s Pod Network
3306                              ALLOW       192.168.1.0/24        # Home LAN
Anywhere (v6) on cni0             ALLOW       Anywhere (v6)
Anywhere (v6) on flannel.1        ALLOW       Anywhere (v6)

mpolak@old-dell:~$ sudo nano /etc/mysql/mariadb.conf.d/50-server.cnf
mpolak@old-dell:~$ sudo vim /etc/mysql/mariadb.conf.d/50-server.cnf
mpolak@old-dell:~$ sudo systemctl restart mariadb
```

```
# https://mariadb.com/kb/en/innodb_buffer_pool_size = 512M
```

Instalacja i Konfiguracja MariaDB na Debianie #4

```
mpolak@old-dell:~$ sudo mysql
[sudo] password for mpolak:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 868
Server version: 10.11.14-MariaDB-0+deb12u2 Debian 12

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE USER IF NOT EXISTS 'tab_view'@'192.168.8.%' IDENTIFIED BY 'tab_view';
Query OK, 0 rows affected (0.054 sec)

MariaDB [(none)]> ALTER USER 'tab_view'@'192.168.8.%' IDENTIFIED BY 'tab_view';
Query OK, 0 rows affected (0.034 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON tab_view.* TO 'tab_view'@'192.168.8.%';
Query OK, 0 rows affected (0.042 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> exit;
Bye
```

Próba połączenia K3s z GPU w celu hostowania modelu LLM #1

Powód: Nvidia Optimus są problematyczne w konfiguracji (kłaster pomimo poprawnych ustawień nie wykrywał plików)

```
🐱> sudo cp /var/lib/rancher/k3s/agent/etc/containerd/config.toml /var/lib/rancher/k3s/agent/etc/containerd/config.toml.tpl  
[sudo] hasło użytkownika mpolak:
```

```
~ took 6s
```

```
🐱> sudo nvidia-ctk runtime configure --runtime=containerd --config=/var/lib/rancher/k3s/agent/etc/containerd/config.toml.tpl  
INFO[0000] Using config version 3  
INFO[0000] Using CRI runtime plugin name "io.containerd.cri.v1.runtime"  
INFO[0000] Wrote updated config to /etc/containerd/conf.d/99-nvidia.toml  
INFO[0000] It is recommended that containerd daemon be restarted.
```

```
🐱> sudo systemctl restart k3s
```

```
~ took 5s
```

```
🐱> kubectl apply -f https://raw.githubusercontent.com/NVIDIA/k8s-device-plugin/v0.15.0/deployments/static/nvidia-device-plugin.yml  
daemonset.apps/nvidia-device-plugin-daemonset created
```

Próba połączenia K3s z GPU w celu hostowania modelu LLM #2

```
~  
🐱> kubectl describe node endeavour | grep -A 5 "Capacity:"  
Capacity:  
  cpu: 16  
  ephemeral-storage: 238865Mi  
  hugepages-1Gi: 0  
  hugepages-2Mi: 0  
  memory: 16045476Ki  
  
~  
🐱> sudo vim /var/lib/rancher/k3s/agent/etc/containerd/config.toml.tpl  
[sudo] hasło użytkownika mpolak:  
  
~ took 1m14s  
🐱> sudo systemctl restart k3s  
  
~ took 5s  
🐱> kubectl delete pod -n kube-system -l name=nvidia-device-plugin-ds  
pod "nvidia-device-plugin-daemonset-4xmt2" deleted from kube-system namespace  
pod "nvidia-device-plugin-daemonset-q2hnm" deleted from kube-system namespace  
pod "nvidia-device-plugin-daemonset-rvfqn" deleted from kube-system namespace  
  
~ took 2s  
🐱> kubectl describe node endeavour | grep -A 5 "Capacity:"  
Capacity:  
  cpu: 16  
  ephemeral-storage: 238865Mi  
  hugepages-1Gi: 0  
  hugepages-2Mi: 0  
  memory: 16045476Ki
```

```
[plugins."io.containerd.grpc.v1.cri".containerd]  
  default_runtime_name = "nvidia"  
[plugins."io.containerd.grpc.v1.cri".containerd.runtimes.nvidia]  
  runtime_type = "io.containerd.runc.v2"  
[plugins."io.containerd.grpc.v1.cri".containerd.runtimes.nvidia.options]  
  BinaryName = "/usr/bin/nvidia-container-runtime"
```

Budowa klastra K3s



kubernetes

CELE

- Instalacja i konfiguracja K3s na wszystkich 3 maszynach
- Przedstawienie działania systemu niemutowalnego (Flatcar)

INSTALACJA K3S NA ENDEAVOUROS

```
🐱> curl -sL https://get.k3s.io | INSTALL_K3S_EXEC="server --node-ip=192.168.8.150 --write-kubeconfig-mode=644" sh -
[INFO] Finding release for channel stable
[INFO] Using v1.35.4+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.35.4%2Bk3s1/sha256sum-amd64.txt
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.35.4%2Bk3s1/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubectl symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Creating /usr/local/bin/ctr symlink to k3s
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s.service
[INFO] systemd: Enabling k3s unit
Created symlink '/etc/systemd/system/multi-user.target.wants/k3s.service' -> '/etc/systemd/system/k3s.service'.
[INFO] systemd: Starting k3s

~ took 11s
🐱> sudo systemctl status k3s
● k3s.service - Lightweight Kubernetes
   Loaded: loaded (/etc/systemd/system/k3s.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-05-15 13:11:20 CEST; 1min 9s ago
  Invocation: e2a038ef0f794840a27361e9ff50c836
     Docs: https://k3s.io
```

```
🐱> kubectl get nodes -o wide
```

NAME	STATUS	ROLES	AGE	VERSION	INTERNAL-IP	EXTERNAL-IP	OS-IMAGE	KERNEL-VERSION	CONTAINER-RUNTIME
endeavour	Ready	control-plane	9m47s	v1.35.4+k3s1	192.168.8.150	<none>	EndeavourOS	7.0.6-arch1-1	containerd://2.2.3-k3s1

INSTALACJA AGENTÓW K3S NA DEBIANIE I FLATCARZE

```
mpolak@old-dell:~$ curl -sL https://get.k3s.io | K3S_URL=https://192.168.8.150:6443 K3S_TOKEN="K10dd956b28c3794114a8fc3e8ad5d04e56d33dc90d1e345d84bf52cec368fb9a11::server:c1ea84b3d3cfec25fcd03f9607ea71bb" sh -
[INFO] Finding release for channel stable
[INFO] Using v1.35.4+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.35.4%2Bk3s1/sha256sum-amd64.txt
[INFO] Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.35.4%2Bk3s1/k3s
[INFO] Verifying binary download
[INFO] Installing k3s to /usr/local/bin/k3s
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /usr/local/bin/kubectl symlink to k3s
[INFO] Creating /usr/local/bin/crictl symlink to k3s
[INFO] Creating /usr/local/bin/ctr symlink to k3s
[INFO] Creating killall script /usr/local/bin/k3s-killall.sh
[INFO] Creating uninstall script /usr/local/bin/k3s-agent-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s-agent.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s-agent.service
[INFO] systemd: Enabling k3s-agent unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s-agent.service → /etc/systemd/system/k3s-agent.service.
[INFO] systemd: Starting k3s-agent
```

```
core@localhost ~ $ curl -sL https://get.k3s.io | INSTALL_K3S_BIN_DIR=/opt/bin K3S_URL=https://192.168.8.150:6443 K3S_TOKEN="K10dd956b28c3794114a8fc3e8ad5d04e56d33dc90d1e345d84bf52cec368fb9a11::server:c1ea84b3d3cfec25fcd03f9607ea71bb" sh -
[INFO] Finding release for channel stable
[INFO] Using v1.35.4+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.35.4%2Bk3s1/sha256sum-amd64.txt
[INFO] Skipping binary downloaded, installed k3s matches hash
[INFO] Finding available k3s-selinux versions
[ERROR] Failed to find the k3s-selinux policy, please install:
  yum install -y container-selinux
  yum install -y https://rpm.rancher.io/k3s/stable/common/centos/8/noarch/k3s-selinux-1.6-1.el8.noarch.rpm
```

```
core@localhost ~ $ yum install -y container-selinux
```

```
-bash: yum: command not found
```

```
core@localhost ~ $ curl -sL https://get.k3s.io | INSTALL_K3S_SKIP_SELINUX_RPM=true INSTALL_K3S_BIN_DIR=/opt/bin K3S_URL=https://192.168.8.150:6443 K3S_TOKEN="K10dd956b28c3794114a8fc3e8ad5d04e56d33dc90d1e345d84bf52cec368fb9a11::server:c1ea84b3d3cfec25fcd03f9607ea71bb" sh -
```

```
[INFO] Finding release for channel stable
[INFO] Using v1.35.4+k3s1 as release
[INFO] Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.35.4%2Bk3s1/sha256sum-amd64.txt
[INFO] Skipping binary downloaded, installed k3s matches hash
[INFO] Skipping installation of SELinux RPM
[INFO] Creating /opt/bin/kubectl symlink to k3s
[INFO] Skipping /opt/bin/crictl symlink to k3s, command exists in PATH at /usr/bin/crictl
[INFO] Skipping /opt/bin/ctr symlink to k3s, command exists in PATH at /usr/bin/ctr
[INFO] Creating killall script /opt/bin/k3s-killall.sh
[INFO] Creating uninstall script /opt/bin/k3s-agent-uninstall.sh
[INFO] env: Creating environment file /etc/systemd/system/k3s-agent.service.env
[INFO] systemd: Creating service file /etc/systemd/system/k3s-agent.service
[INFO] systemd: Enabling k3s-agent unit
Created symlink '/etc/systemd/system/multi-user.target.wants/k3s-agent.service' → '/etc/systemd/system/k3s-agent.service'.
[INFO] systemd: Starting k3s-agent
```

SPRAWDZENIE CZY CONTROL PLANE WYKRYWA WSZYSTKIE WORKER NODY + NAŁOŻENIE WYJĄTKU DLA DEBIANA (PRZEZ MARIADB)

```
🐱 ) kubectl get nodes -o wide
```

NAME	STATUS	ROLES	AGE	VERSION	INTERNAL-IP	EXTERNAL-IP	OS-IMAGE	KERNEL-VERSION	CONTAINER-RUNTIME
endeavour	Ready	control-plane	41m	v1.35.4+k3s1	192.168.8.150	<none>	EndeavourOS	7.0.6-arch1-1	containerd://2.2.3-k3s1
localhost	Ready	<none>	3m2s	v1.35.4+k3s1	192.168.8.170	<none>	Flatcar Container Linux by Kinvolk 4593.2.1 (Oklo)	6.12.87-flatcar	containerd://2.2.3-k3s1
old-dell	Ready	<none>	3m2s	v1.35.4+k3s1	192.168.8.160	<none>	Debian GNU/Linux 12 (bookworm)	6.1.0-45-amd64	containerd://2.2.3-k3s1

```
🐱 ) kubectl taint nodes old-dell database=true:NoSchedule  
node/old-dell tainted
```

Wdrożenie aplikacji na K3s

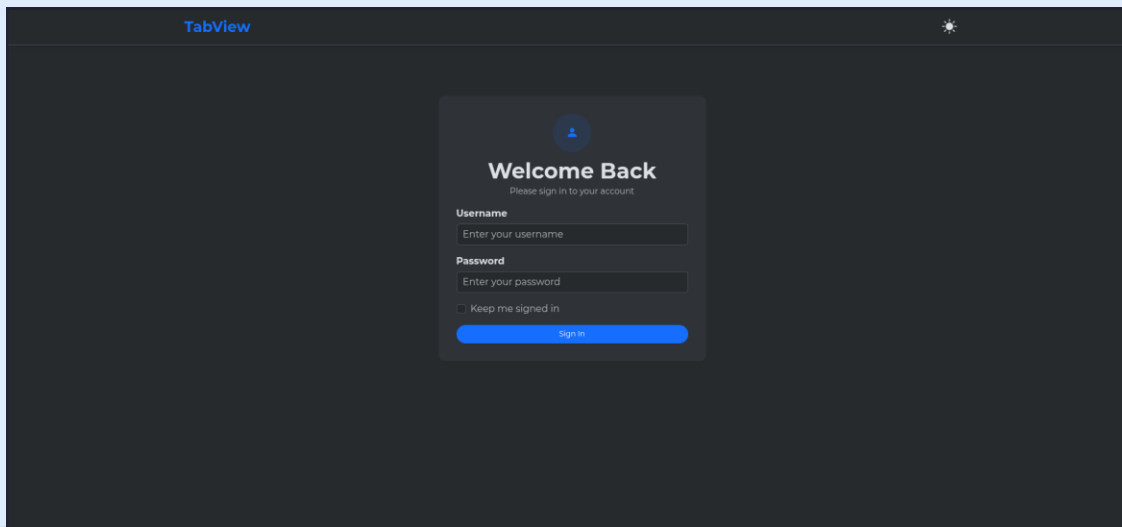


CELE

- Napisanie Manifestu w Yaml
- Postawienie całej aplikacji z deklarowanymi obiektami K3s (Namespace, ConfigMap, Secret, PersistentVolumeClaim, Deployment, Service, Ingress, NetworkPolicy, HorizontalPodAutoscaler)
- Sprawdzenie czy aplikacja działa i można do niej wejść
- Przetestowanie działania HPA (replikacja Podów)

PROCES WDROŻENIA APLIKACJI

```
~/Pulpit/Projekt Kubernetes took 46s
❯ kubectl apply -f k3s.yaml
namespace/prod created
configmap/tabview-config created
secret/tabview-secret created
persistentvolumeclaim/tabview-uploads-pvc created
deployment.apps/tabview-deployment created
service/tabview-service created
ingress.networking.k8s.io/tabview-ingress created
horizontalpodautoscaler.autoscaling/tabview-hpa created
networkpolicy.networking.k8s.io/tabview-ingress-policy created
```



```
apiVersion: v1
kind: Namespace
metadata:
  name: prod
  labels:
    environment: production
    app: tabview
    namespace: "Namespace for Tabview app"
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: tabview-config
  namespace: prod
  labels:
    app: tabview
data:
  DB_URL: "postgres://db:5432"
  PLAIN_APP: "tabview"
  PLAIN_SECRET: ""
  AUTH_SECRET_STORAGE_URL: "memory://"
---
apiVersion: v1
kind: Secret
metadata:
  name: tabview-secret
  namespace: prod
  labels:
    app: tabview
type: Opaque
data:
  SECRET_KEY: "abcdefghijklmnopqrstuvwxyz0123456789"
  DB_PLAIN_APP: "tabview"
  DB_PLAIN_SECRET: ""
  AUTH_SECRET_STORAGE_URL: "memory://"
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: tabview-uploads-pvc
  namespace: prod
spec:
  storageClassName: standard
  resources:
    requests:
      storage: 1Gi
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: tabview-deployment
  namespace: prod
spec:
  replicas: 1
  selector:
    matchLabels:
      app: tabview
  template:
    metadata:
      labels:
        app: tabview
    spec:
      containers:
        - name: tabview-app
          image:
            name: tabview-app
            imagePullPolicy: Always
            ports:
              - containerPort: 8080
            envFrom:
              - configMapRef:
                  name: tabview-config
              - secretRef:
                  name: tabview-secret
            resources:
              requests:
                memory: "128Mi"
                cpu: "100m"
              limits:
                memory: "128Mi"
                cpu: "100m"
            volumeMounts:
              - path: /health
                mountPath: /opt/app
                subPath: ""
              - path: /health
                mountPath: /opt/app
                subPath: ""
              - name: uploads-storage
                mountPath: /app/tabview/uploads
            volume:
              name: uploads-storage
              persistentVolumeClaim:
                claimName: tabview-uploads-pvc
---
apiVersion: v1
kind: Service
metadata:
  name: tabview-service
  namespace: prod
spec:
  selector:
    app: tabview
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
---
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: tabview-ingress
  namespace: prod
  labels:
    app: tabview
spec:
  rules:
    - host:
        path: /
        pathType: Prefix
      http:
        paths:
          - name: tabview-service
            path: /
            backend:
              serviceName: tabview-service
              port:
                number: 80
---
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: tabview-hpa
  namespace: prod
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: tabview-deployment
  minReplicas: 1
  maxReplicas: 1
  metrics:
    - type: Resource
      resource:
        name: cpu
      target:
        type: Utilization
        averageUtilization: 80
---
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: tabview-ingress-policy
  namespace: prod
spec:
  podSelector:
    matchLabels:
      app: tabview
  ingress:
    - from:
        - namespaceSelector:
            namespaceNames:
              - kube-system
      ports:
        - protocol: TCP
          port: 8080
```

Repo z manifestami:

<https://github.com/MaxPolak04/k3s-tabview-infra>

TEST HPA -> ZWIĘKSZENIE LICZBY PODÓW PRZEZ PRZECIĄŻENIE #1

```
🐧 ) for i in {1..20}; do while true; do curl -s http://192.168.8.150 > /dev/null; done & done  
[2] 50748  
[3] 50750  
[4] 50752  
[5] 50753  
[6] 50756  
[7] 50758  
[8] 50760  
[9] 50761  
[10] 50764  
[11] 50766  
[12] 50767  
[13] 50769  
[14] 50771  
[15] 50773  
[16] 50775  
[17] 50777  
[18] 50780  
[19] 50781  
[20] 50784  
[21] 50785
```

TEST HPA -> ZWIĘKSZENIE LICZBY PODÓW PRZEZ PRZECIĄŻENIE #2

```
Co 2,0s: kubectl get hpa,pods -n prod                                     Endeavour: nie, 17 maj 2026, 17:56:22
                                                                           in 0,532s (0)
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS
horizontalpodautoscaler.autoscaling/tabview-hpa	Deployment/tabview-deployment	cpu: 1%/10%	2	5

NAME	READY	STATUS	RESTARTS	AGE
pod/tabview-deployment-769cdf8956-cfkmf	0/1	CrashLoopBackOff	2 (22s ago)	73s
pod/tabview-deployment-769cdf8956-jtfhk	1/1	Running	17 (11m ago)	53m
pod/tabview-deployment-769cdf8956-n2kgk	1/1	Running	0	4m42s

```
Co 2,0s: kubectl get hpa,pods -n prod                                     Endeavour: nie, 17 maj 2026, 17:57:40
                                                                           in 0,120s (0)
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
horizontalpodautoscaler.autoscaling/tabview-hpa	Deployment/tabview-deployment	cpu: 483%/10%	2	5	5	55m

NAME	READY	STATUS	RESTARTS	AGE
pod/tabview-deployment-769cdf8956-62j5w	0/1	Running	0	30s
pod/tabview-deployment-769cdf8956-cfkmf	1/1	Running	3 (100s ago)	2m31s
pod/tabview-deployment-769cdf8956-jtfhk	1/1	Running	17 (12m ago)	55m
pod/tabview-deployment-769cdf8956-n2kgk	1/1	Running	0	6m
pod/tabview-deployment-769cdf8956-zlh9b	0/1	Running	0	30s

Wdrożenie ArgoCD

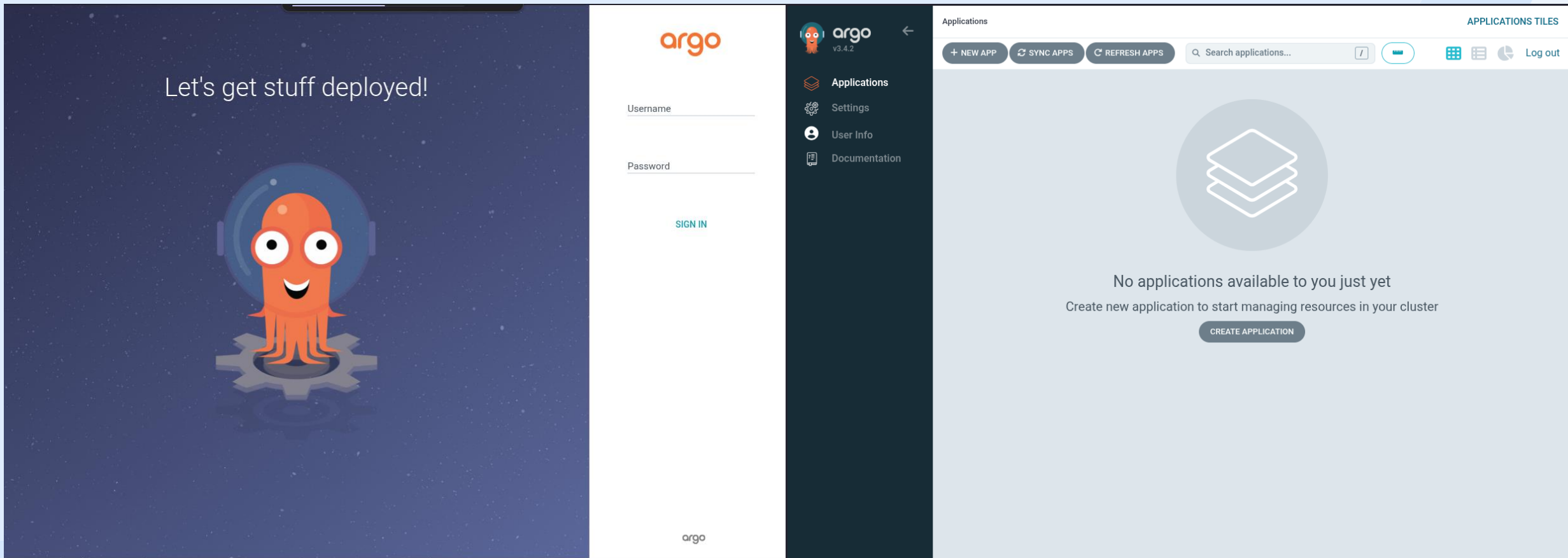
CELE

- Instalacja ArgoCD
- Przekierowanie portów w celu wejścia do dashboardu
- Wydobucie hasła admina
- Logowanie, zmiana hasła admina i rozeznanie w dashboardzie
- Podzielenie manifestu na mniejsze części
- Utworzenie repo z manifestami i podłączenie do ArgoCD
- Przetestowanie ArgoCD (Single Source of Truth) poprzez zmianę K3s (usunięcie Deploymentu)

INSTALACJA ARGOCD

[illegible]

LOGOWANIE DO ARGOCD



SYNCHRONIZACJA Z REPO

MaxPolak04

Add manifests

62caefa · 1 minute ago

manifests

Add manifests

1 minute ago

LICENSE

Initial commit

38 minutes ago

README.md

Add manifests

1 minute ago

README

MIT license

Hybrid K3s Cluster - Infrastructure Specification

This repository serves as the central **Single Source of Truth** for the declarative configuration of resources within a hybrid Kubernetes (K3s) cluster. The project implements a modern **Infrastructure as Code (IaC)** approach, completely separating the application source code from its runtime environment definitions.

The infrastructure is designed to simulate professional **Cloud Native** operations, deployment automation, and advanced Linux systems administration.

argo

v3.4.2

Applications

Settings

User Info

Documentation

Resource filters

NAME

KINDS

SYNC STATUS

HEALTH STATUS

tabview-production

5 minutes

prod

tabview-uploads-pvc

tabview-secret

tabview-service

tabview-deployment

tabview-hpa

tabview-ingress

tabview-ingress-policy

tabview-production

5 minutes

prod

tabview-uploads-pvc

tabview-secret

tabview-service

tabview-deployment

tabview-hpa

tabview-ingress

tabview-ingress-policy

tabview-deployment-769cdf8...

2 hours

rev:1

tabview-deployment-769cdf8...

2 hours

rev:1

tabview-deployment-769cdf8...

2 hours

running

1/1

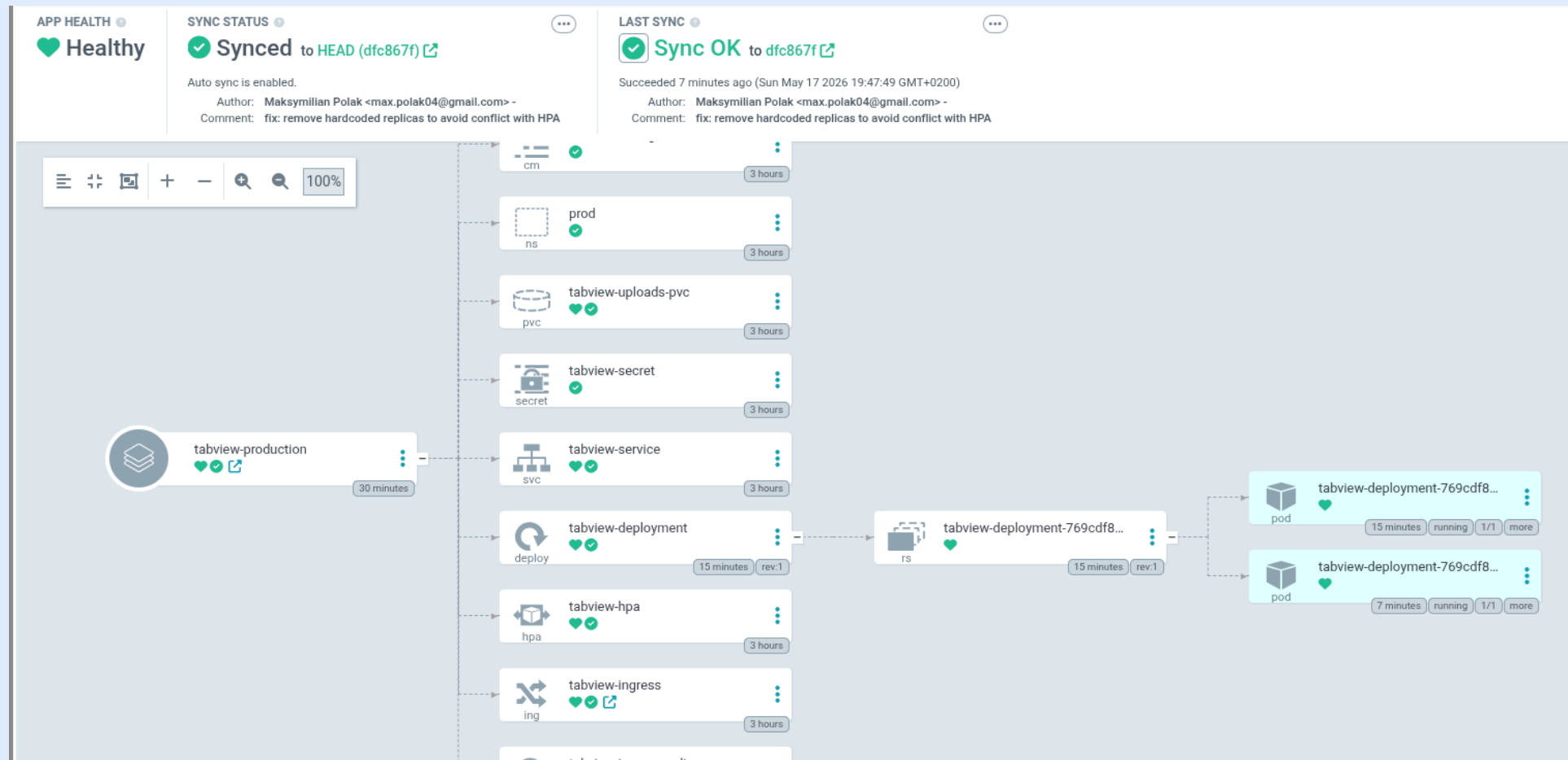
more

TEST ARGOCD

```
➜ sudo k3s kubectl delete deployment tabview-deployment -n prod
[sudo] hasło użytkownika mpolak:
deployment.apps "tabview-deployment" deleted from prod namespace
```

The screenshot displays the Argo CD web interface for the 'tabview-production' application. The interface is in a dark theme. The left sidebar shows navigation options: Applications, Settings, User Info, and Documentation. The main content area displays the application's health status as 'Degraded' and its sync status as 'OutOfSync'. The 'APPLICATION DETAILS TREE' on the right lists various Kubernetes resources including namespaces, PVCs, secrets, services, deployments, HPAs, ingress, and policies. The bottom status bar shows the time as 19:34 on 17.05.2026.

ARGOCD SAMO SIĘ SYNCHRONIZUJE Z REPO I ODZWIERCIEDLA JEGO STAN



Dziękujemy za uwagę

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